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Fall 2024 Applied PDE's 440/540 Schreyer

TEST 2

1. Please PRINT your name at the top of this page.
2. Check to see that you have 9 pages and 6 problems. The last 2 pages are blank for scratch work.
3. If you do your final work on another page, please indicate clearly where you do your work.
4. One side of an 8.5" by 11" sheet of paper is allowed for notes (no examples). Please turn this page in with the exam.
5. No technology.
6. If you run out of time, explain what you would do if you did have time.
7. If you need a formula that you do not have on your note sheet, you may ask for it and if you receive it, 2 points will be deducted from that question.
8. Show what you know.

Page 2 15 (18)

Page 3 14 (16)

Page 4 18 (18)

Page 5 10 (10)

Page 6 10 (10)

Page 7 10 (10)

Total 77 (~~72~~ 82)

1. (18 pts) Short answer

6

(a) List three criteria that makes a PDE problem well-posed.

A PDE is called well-posed if

① There exist a solution,

② The solution is unique

③ The solution continously depends on data (IC, BC's) i.e. with IC or BC's defined by $f(\underline{x})$, $|f(\underline{x}) - \tilde{f}(\underline{x})| < \delta \Rightarrow |u - \tilde{u}| < \varepsilon(\delta)$.

4

(b) Write $\nabla u \cdot \nabla u$ in $\mathbb{R}^3$ with independent variables x, y, z.

$\nabla u = \langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \rangle$, so

$\nabla \cdot u(x,y,z) \cdot \nabla u(x,y,z) = \langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \rangle \cdot \langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \rangle =$ ← Dot Product

$\neq \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$.

be careful

5

(c) Write down a PDE problem (i.e. the PDE and the boundary condition and where each of these hold) that describes the following two physical problems. DO NOT SOLVE. It is not necessary to write down how smooth the solution must be.

$u_t = \Delta u \Rightarrow$ steady-state temp. distr $\Rightarrow$ $\boxed{\Delta u = 0}$

Find the steady-state temperature distribution of a two-dimensional disk (please ask me if you do not know what a "disk" is) with radius 2, where on the boundary

i. the temperature is equal to $3\sin(\theta)$

ii. the boundary is insulated.

Ω = disk of radius 2 $\Rightarrow$

$\Omega = \{(r,\theta) : 0 < r < 2,\ 0 \le \theta \le 2\pi\}$

Note open

Sol-n:

$\Delta u = 0$, Ω

(i) $u(r,\theta;t) = 3\sin(\theta)$, $\partial\Omega$

(ii) $\frac{\partial u}{\partial \underline{n}} = 0$, $\partial\Omega$

what is $\partial\Omega$?

Where the Laplace operator, Δu, on polar coordinates is as follows:

$$\Delta u = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \cdot \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \cdot \left[\frac{\partial^2 u}{\partial \theta^2}\right].$$

not sure about the "order", $|\alpha|$, so am I supposed to iterate α symbols until we get 6 nonzero terms in the Taylor expansion formula?

14

2. (16 pts) Find the first 6 nonzero terms of the Taylor series for the function:

$$(1) \qquad f(x,y) = e^{5x+3y}$$

about the origin, $(0,0)$. You must use the multi-dimensional version of Taylor series. No points for not using the multi-dimensional form of the Taylor series.

Sol-n: Let

(1) $f(x,y) = f(\underline{x}) = e^{5x+3y}$

(2) $\underline{x}^0 = (0,0)$ the expansion pt. Then

(3) $$T_n(\underline{x}) = \sum_{0 \le |\alpha| \le n} \frac{D^\alpha f(\underline{x}^0)}{\alpha!} \cdot (\underline{x} - \underline{x}^0)^\alpha$$ where the symbol α i

$\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$.

Since I don't know exactly what order ($n = ?$) gives us the first 6 nonzero terms, let me start directly computing

(4) $\boxed{\frac{D^\alpha f(\underline{x}^0)}{\alpha!} \cdot (\underline{x} - \underline{x}^0)^\alpha}$ terms until we get the first 6 nonzero of these:

Let

(5) $\alpha = \underset{a)}{(0,0)}, \underset{b)}{(1,0)}, \underset{c)}{(0,1)}, \underset{d)}{(2,0)}, \underset{e)}{(1,1)}, \underset{f)}{(0,2)},$ $\underset{g)}{(3,0)}, \underset{h)}{(2,1)}, \underset{i)}{(1,2)}, \underset{j)}{(0,3)}, \ldots$ if necessary

a) $\alpha = (0,0)$

(6) $$\frac{D^{(0,0)} f(\underline{x})}{(0,0)!}\Bigg|_{(0,0)} \overbrace{(\underline{x}-\underline{0})^{(0,0)}}^{=1} = \frac{1}{0!\cdot 0!} \cdot \overbrace{\frac{\partial^{0+0} f(x,y)}{\partial x^0\, \partial y^0}}^{\text{zero partials} = f(x,y)}\Bigg|_{(0,0)} = \frac{1}{1} \cdot \left[e^{5x+3y}\right]_{(0,0)}$$

$= 1 \cdot e^{0+0} = 1 \neq 0$ ✓ (first nonzero term)

3 1

! as usual. Thank you - blue sheet

$\eta = x - 3\xi = x - 3(y - x) = 4x - 3y$

18

3. (18 pts) Consider the following PDE:

(1)

$$\frac{\partial^2 u}{\partial x^2} + 4u = e^y.$$

3 (a) (3 pts) Classify the equation as elliptic, hyperbolic, or parabolic.

4 (b) (4 pts) Consider the change of variables

$$x = 3\xi + \eta, \qquad y = 4\xi + \eta$$
$$\xi = -x + y, \qquad \eta = 4x - 3y$$

Is the transformation invertible? Check Jacobian.

8 (c) (8 pts) Transform the PDE above from $u(x, y)$ to $u(\xi, \eta)$. Note: The resulting PDE should have no x nor y in the equation.

3 (d) (3 pts) Classify the PDE in the (ξ, η) coordinate system.

Soln: [Preamble]

(a) So the given eq-n (1) is

$$\underbrace{1}_{a} \cdot u_{xx} + 4u - e^y = 0$$

Principal part: $1 \cdot u_{xx}$, then $a = 1$, $b = 0$, $c = 0$

$\Delta = b^2 - 4ac = 0 - 4 \cdot 1 \cdot 0 = 0 \Rightarrow$ (1) is parabolic PDE.

(b) $\underbrace{\begin{cases} x = 3\xi + \eta \\ y = 4\xi + \eta \end{cases}}_{(3)}$ - $\underbrace{\begin{cases} \xi = y - x \\ \eta = 4x - 3y \end{cases}}_{(2)}$ $\Rightarrow$ being able to find $(x, y) \rightleftarrows (\xi, \eta)$, so invertible. More clearly,

(4) By (2) $J = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} = \begin{vmatrix} -1 & 1 \\ 4 & -3 \end{vmatrix} = 3 - 4 = -1 \neq 0 \Rightarrow$

thus the given trans-n (2) + (3) is invertible.

(c)

(5) — $u_x = u_\xi \xi_x + u_\eta \eta_x$ (Total deriv. by chain rule) &

(6) — — $u_{xx} = u_{\xi\xi} \xi_x^2 + 2u_{\xi\eta} \xi_x \eta_x + u_{\eta\eta} (\eta_x)^2 + u_\xi \xi_{xx} + u_\eta \eta_{xx}$ because

$\rightarrow$ 4,

b) $\alpha = (1,0)$

$\langle x, y \rangle$ always do next time, skipp

(7) $$\frac{D^{(1,0)} f(\underline{x})}{(1,0)!}\Bigg|_{(0,0)} (\underline{x} - \underline{0})^{(1,0)} = \frac{\partial^{1+0} f(x,y)}{\partial x^1 \partial y^0}\Bigg|_{(0,0)} \overbrace{(\langle x,y\rangle - \langle 0,0\rangle)}^{\parallel}{}^{(1,0)} =$$

$$= \frac{\partial f}{\partial x}\Bigg|_{(0,0)} \cdot (x,y)^{(1,0)} = \frac{\partial}{\partial x}\left[e^{5x+3y}\right]_{(0,0)} \cdot x^1 \cdot y^0 =$$

$$= 5 \cdot \left[e^{5x+3y}\right]_{(0,0)} \cdot x = 5x \neq 0 \quad (2^{nd} \text{ one})$$

(c) $\alpha = (0,1)$

(8) $$\frac{D^{(0,1)} f(\underline{x})}{(0,1)!}\Bigg|_{(0,0)} (\underline{x} - \underline{0})^{(0,1)} = \frac{\partial^{0+1} f(x,y)}{\partial x^0 \partial y^1}\Bigg|_{(0,0)} \cdot (x,y)^{(0,1)} =$$

$$= \frac{\partial f}{\partial y}\Bigg|_{(0,0)} \cdot x^0 \cdot y^1 = \frac{\partial}{\partial y}\left[e^{5x+3y}\right]_{(0,0)} \cdot y = 3 \cdot \left[e^{5x+3y}\right]_{(0,0)} \cdot y$$

$$= 3 \cdot e^0 \cdot y = 3y \neq 0 \quad (3^{rd} \text{ one})$$

(d) $\alpha = (2,0)$, $|\alpha| = 2$

(9) $$\frac{D^{(2,0)} f(\underline{x})}{(2,0)!}\Bigg|_{(0,0)} \cdot (\underline{x} - \underline{0})^{(2,0)} = \frac{\partial^2 f(x,y)}{\partial x^2}\Bigg|_{(0,0)} \cdot x^2 =$$

by (7)

$$= \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right]_{(0,0)} \cdot x^2 = \frac{\partial}{\partial x}\left[5 \cdot e^{5x+3y}\right]_{(0,0)} \cdot x^2 = 25\left[e^{5x+3y}\right]_{(0,0)} \cdot x^2 =$$

$$= 25 \cdot e^0 \cdot x^2 = 25x^2 \neq 0 \quad (4^{th} \text{ one})$$

(e) $\alpha = (1,1)$, $|\alpha| = 2$ missed this page

(10) $$\frac{D^{(1,1)} f(\underline{x})}{(1,1)!}\Big|_{(0,0)} \cdot (\underline{x} - \underline{0})^{(1,1)} = \frac{\partial^{1+1} f(x,y)}{\partial x^1 \partial y^1}\Big|_{(0,0)} (x,y)^{(1,1)} =$$

by (8)

$$= \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right]_{(0,0)} x^1 \cdot y^1 = \frac{\partial}{\partial x}\left[3e^{5x+3y}\right]_{(0,0)} \cdot xy =$$

$$= 15 \cdot \left[e^{5x+3y}\right]_{(0,0)} \cdot xy = 15 \cdot e^0 \cdot xy = 15xy \neq 0 \text{ (5th one)}$$

(f) $\alpha = (0,2)$ $|\alpha| = 2 \Rightarrow \alpha! = (0!)(2!) = (1)(2) = 2$

(11) $$\frac{D^{(0,2)} f(\underline{x})}{(0,2)!} \cdot (\underline{x} - \underline{0})^{(0,2)} = \frac{\partial^{0+2} f(x,y)}{\partial x^0 \partial y^2}\Big|_{(0,0)} \cdot (x,y)^{(0,2)} =$$

$$= \frac{\partial^2 f}{\partial y^2}\Big|_{(0,0)} x^0 \cdot y^2 = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial y}\right]_{(0,0)} y^2 = \frac{\partial}{\partial y}\left[3 \cdot e^{5x+3y}\right]_{(0,0)} \cdot y^2 =$$

by (8)

$$= 9 \cdot \left[e^{5x+3y}\right]_{(0,0)} y^2 = 9 \cdot e^0 \cdot y^2 = 9y^2 \neq 0 \text{ (6th one)}$$

So we first 6 non zero terms have obtained in the 2nd order, $n=2$, $|\alpha| \leq 2$. So the (3) Taylor formula can be written as follows: (Σ results in (6),... (11).)

$$T_2(x) = \sum_{0 \leq |\alpha| \leq 2} \frac{D^\alpha f(\underline{0})}{\alpha!} (\underline{x} - \underline{0})^\alpha =$$

$$= 1 + 5x + 3y + 25x^2 + 15xy + 9y^2.$$

$$u_{xx} = \frac{\partial}{\partial x}\left[u_x\right] = \frac{\partial}{\partial x}\left[u_\xi \xi_x + u_\eta \eta_x\right] = \frac{\partial}{\partial x}\left[u_\xi \xi_x\right] + \frac{\partial}{\partial x}\left[u_\eta \eta_x\right] =$$

$$= (u_{\xi\xi}\xi_x + \underline{u_{\xi\eta}}\eta_x)\cdot\xi_x + (\underline{u_{\eta\xi}}\xi_x + u_{\eta\eta}\eta_x)\cdot\eta_x + u_\xi \xi_{xx} + u_\eta \eta_{xx} =$$

$$= u_{\xi\xi}\xi_x^2 + 2u_{\xi\eta}\xi_x\eta_x + u_{\eta\eta}\eta_x^2 + u_\xi\xi_{xx} + u_\eta\eta_{xx}. \quad \text{So}$$

(7) $\begin{cases} \xi_x = -1 & \xi_{xx} = 0 \\ \eta_x = 4 & \eta_{xx} = 0 \end{cases}$, then

$$u_{xx} = u_{\xi\xi}\cdot(-1)^2 + 2u_{\xi\eta}\cdot(-1)\cdot(4) + u_{\eta\eta}(4)^2 + u_\xi\cdot 0 + u_\eta\cdot 0 \Rightarrow$$

(8) $u_{xx} = u_{\xi\xi} - 8u_{\xi\eta} + 16u_{\eta\eta}$

(9) $u(x,y) = u(\xi,\eta)$, & finally $y = 4\xi + \eta$ so

(10) $e^y = e^{4\xi+\eta}$. Putting all (8), (9), (10) together into (1) gives

$$u_{xx} + 4u - e^y = \overbrace{u_{\xi\xi} - 8u_{\xi\eta} + 16u_{\eta\eta}}^{u_{xx}} + 4\overbrace{u}^{u} - \overbrace{e^{4\xi+\eta}}^{e^y} = 0 \Rightarrow$$

(11) $\underbrace{u_{\xi\xi} - 8u_{\xi\eta} + 16u_{\eta\eta}} + 4u - e^{4\xi+\eta} = 0.$

(d) Note to check principal part

$u_{\xi\xi} - 8u_{\xi\eta} + 16$ $\quad A = 1,\ B = -8,\ C = 16 \Rightarrow$

$\Delta' = B^2 - 4AC = (-8)^2 - 4\cdot(1)\cdot(16) = 0$ ✓ $\Rightarrow$ good

parabolic, as expected.

$\Delta' = \underbrace{\Delta}_{0}\cdot J^2 \Rightarrow \Delta' = 0.$ ✓

I was not looking at d) but did already long if misorganized.

10

4. (10 pts) Show that the following function is harmonic and describe the domain over which it is harmonic:

(1)
$$u(r,\theta) = \frac{\sin(\theta)}{r}$$

Sol-n. By the def-n of harmonic f-n, (i) $u \in C^2(\Omega)$ (indeed $u \in C^0(\Omega)$ is enough by property 1 of har-c f-ns), (ii) $\Delta u = 0$.

let me check:

(i) $u(r,\theta) = \underbrace{\frac{1}{r}}_{\in C^\infty(\mathbb{R}^2 \setminus \{\underline{0}\})} \cdot \sin(\theta) \in C^\infty(\mathbb{R}^2)$

!

Indeed thought, we're in Polar so $r>0$, but not explicitly given so assuming $\boxed{r<0}$ & $r>0$ like yesterday's HW. otherwise could exclude $r<0$ case.

ok

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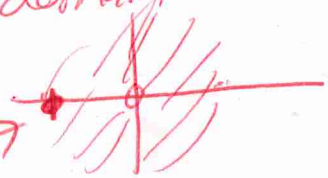
Think of sketch of domain

$r=2, \theta=\pi$

& $r=-2, \theta=0$



then $u = \frac{1}{r} \cdot \sin(\theta) \in C^\infty(\mathbb{R}^2 \setminus \{\underline{0}\})$ ✓

(ii) By Laplace oper- in in polar

$$\Delta u = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \cdot \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \cdot \frac{\partial^2 u}{\partial \theta^2} = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \cdot \frac{\partial}{\partial r}\left[\frac{1}{r} \cdot \sin\theta\right]\right] +$$

$$+ \frac{1}{r^2} \cdot \frac{\partial}{\partial \theta}\left[\frac{\partial}{\partial \theta}\left[\frac{1}{r} \cdot \sin\theta\right]\right] = \frac{1}{r} \cdot \frac{\partial}{\partial r} \cdot \left[r \cdot \left(-\frac{\sin\theta}{r^2}\right)\right] + \frac{1}{r^2} \cdot \frac{\partial}{\partial \theta}\left[\frac{1}{r}\cos\theta\right]$$

$$= \frac{1}{r} \cdot \left[1 \cdot \left(-\frac{\sin\theta}{r^2}\right) + r \cdot \frac{2\sin\theta}{r^3}\right] - \frac{1}{r^2} \cdot \frac{1}{r} \cdot (-\sin\theta) =$$

10

(1)

5. (10 pts) Consider the following equation in $\mathbb{R}^2$:

$$u_{xx} + u_{yy} + u_y = 0$$

Use separation of variables in Cartesian coordinates, (x, y), to get two ordinary differential equations, each with only one dependent variable. Be sure to put the ODE equations in a form traditionally needed when solving an ODE. DO NOT SOLVE THE ODE's. Circle the two equations and go to the next problem.

Soln. [Dr J]

(1) $u(x,y) = X(x) \cdot Y(y) \neq 0$, for some nonzero $X \neq 0$, $Y \neq 0$ $\Rightarrow$ (1) is if $X=0$ or $Y=0$ (1) is trivial.

good

$$Y X'' + X Y'' + X \cdot Y' = 0.$$

$$\frac{X'' \cdot Y}{XY} + \frac{X \cdot Y''}{YX} + \frac{X \cdot Y'}{XY} = 0 \quad \Big| : XY \neq 0$$

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Y'}{Y} = 0.$$

$$-\frac{X''}{X} = \frac{Y'' + Y'}{Y} = \mu.$$

(1) $-\frac{X''}{X} = \mu \Rightarrow -X'' - X\mu = 0$

(2) $\frac{Y'' + Y'}{Y} = \mu \Rightarrow Y'' + Y' - \mu Y = 0.$

good!

$$= \frac{-\sin\theta}{r^3} + \frac{2\sin\theta}{r^3} - \frac{\sin\theta}{r^3} = \frac{2\sin\theta - \sin\theta - \sin\theta}{r^3} = \frac{0}{r^3} \equiv 0,$$

that is

$$\Delta u \equiv 0$$

hence u is harmonic. $\blacksquare$

(Literally running out of time!)

10 6. (10 pts) In the process of using separation of variables for a PDE, we assume a solution of the form $u(x, y) = X(x)Y(y)$. The following two ODE's arise:

$$X'' + \mu X = 0 \qquad Y' + \mu Y = 0.$$

(1) (2)

(a) where $X(x)$ and $Y(y)$. Note that the second equation is a first-order differential equation. Find 2 linearly independent solutions for $u(x, y)$ assuming that $\mu > 0$.

(b) If you have time, you may want to check your solutions. The original PDE is:

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial y} = 0. \quad (\star)$$

Sol-n. [Preamble] I know hopefull can do it, but might be picking my work!

let me start with (2): $\boxed{\mu > 0}$ (3)

(2) $Y' = -\mu Y \Rightarrow$

$$\frac{dY}{dy} = -\mu Y \qquad \text{(Sep-n of var-s)}$$

$$\int \frac{dY}{Y} = \int -\mu \, dy + C.$$

Assuming all nice $C, D > 0$

$$\ln|Y| = -\mu y + \ln D$$

$$\ln(y) = \ln(e^{-\mu y} \cdot D) \Rightarrow$$

(b) $\boxed{y = e^{-\mu y} \cdot D}$

$$F.S.S(Y) = \{e^{-\mu y}\}$$

Now for (1)

$X'' + \mu X = 0$. Assume $X = e^{\lambda x}$ then $\Rightarrow$

$\lambda^2 + \mu = 0$

$\lambda = \pm\sqrt{\mu}\, i \Rightarrow$

(5) $\boxed{X(x) = C_1 \cos(\sqrt{\mu}\, x) + C_2 \sin(\sqrt{\mu}\, x)} \Rightarrow$

$F.S.S(X) = \{ \cos(\sqrt{\mu}\, x), \sin(\sqrt{\mu}\, x) \} \Rightarrow$

as $u = X \cdot Y$ then,

good!

$F.S.S(u) = \{ \cos(\sqrt{\mu}\, x) \cdot e^{-\mu y}, \sin(\sqrt{\mu}\, x) \cdot e^{-\mu y} \}$.

(b)

For (*), I need u_{xx} u_y = ? (why not asking Laplace wanting harmonic?)

$u_x = \frac{\partial}{\partial x}\left[C_1 \cos(\sqrt{\mu}\, x) e^{-\mu y} + C_2 \sin(\sqrt{\mu}\, x) \cdot e^{-\mu y} \right] =$

$= -C_1 \sqrt{\mu} \cdot \sin(\sqrt{\mu}\, x) e^{-\mu y} + \sqrt{\mu} \cdot C_2 \cos(\sqrt{\mu}\, x) \cdot e^{-\mu y}$ &

$u_y =$ . . .

u_{xy} (seems need to subit-n no time!)

6.) $\alpha = (1,0)$

(7) $$\left.\frac{D^{(1,0)} f(\underline{x})}{(1,0)!}\right|_{(0,0)} = \left.\frac{1}{1!\,0!}\frac{\partial^{1+0} f}{\partial x^1 \partial y^0}\right|_{(0,0)} = 1 \cdot \left.\frac{\partial f}{\partial x}\right|_{(0,0)} = \frac{\partial}{\partial x}\left[e^{5x+3y}\right]_{(0,0)}$$

$$= 5 \cdot \left[e^{5x+3y}\right]_{(0,0)} = 5 \cdot e^0 = 5 \neq 1 \quad (2^{nd}\ \text{ord})$$

$$\frac{\partial}{\partial r}\left[\frac{-\sin\theta}{r^2}\right] = \frac{-\sin\theta \cdot (-2)}{r^3}$$

• Laplace oper-r: $\Delta u = \underline{\nabla}\cdot\underline{\nabla} u = \frac{\partial^2}{\partial x_1^2} + \dots + \frac{\partial^2 u}{\partial x_n^2}$

• Def'n (Harmonic): u is harmonic in the dom-n Ω, if ① $u \in C^2(\Omega)$ & satisfies the Laplace eq-n $\Delta u = 0$ in Ω.

• Thm 7.1 (w/s) Recall Analytic f-ns $\in C^\infty$

If $u \in C^0(\Omega)$ & $\Delta u = 0$, then u is <u>analytic</u> ($\exists$ Taylor exp. $\Rightarrow$ f-n u.) (So Harmonic f-ns are analytic & so smooth $\in C^\infty$).

$C^0(\Omega) = \{f(x) : f(x)$ is cont-s over $\Omega\}$

$C^1(\Omega) = \{f(x) : f(x) \in C^0(\Omega),\ f_{x_i} \in C^0(\Omega)\}$

$C^2(\Omega) = \{f(x) : f(x) \in C^0(\Omega),\ f_{x_i} \in C^0(\Omega),\ f_{x_i x_j} \in C^0(\Omega)\} \dots$

Taylor Series: Notation:

$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ symbol

$$D^\alpha f := D_1^{\alpha_1} D_2^{\alpha_2} \cdots D_n^{\alpha_n} f = \frac{\partial^{\alpha_1+\alpha_2+\dots+\alpha_n} f}{\partial x_1^{\alpha_1} \cdot \partial x_2^{\alpha_2} \cdots \partial x_n^{\alpha_n}}$$

wrt 2nd indep var.

$x^\alpha := x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdots x_n^{\alpha_n}$

$\alpha! := \alpha_1! \cdot \alpha_2! \cdots \alpha_n!$

$|\alpha| := \alpha_1 + \dots \alpha_n$

$$T_n(\underline{x}) = \sum_{0 \le |\alpha| \le n} \frac{D^\alpha f(\underline{x}^0)}{\alpha!} (\underline{x} - \underline{x}^0)^\alpha$$

Taylor Series

$T_{cc} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Higher Order PDE:

(1) $P(\underline{x}, D) = \sum_{|\alpha| \le m} a^\alpha D^\alpha$ = Linear operator (1)

(2) $P_m(\underline{x}, D) = \sum_{|\alpha| \le m} a^\alpha(\underline{x}) D^\alpha$ = Principal part of Lin. op. (1).

(3) $P_m(\underline{x}, \underline{\xi}) = \sum_{|\alpha| \le m} a^\alpha(\underline{x}) \xi^\alpha$ = Principal symbol.

Idea: Info of sol-n of PDE is spread throug a surface $\perp$ ξ. = <u>Characteristic direction</u>

(3) = 0 i.e. $P_m(\underline{x}, \xi) = 0$ solve for ξ. Find vectors $\langle v, \xi \rangle = 0$ for this surface.

---

given $a u_{xx} + b u_{xy} + c u_{yy} + \dots = 0$. $\Delta = b^2 - 4ac$ with $y \ne 0$

$\xi = \xi(x,y)$, $\eta = \eta(x,y)$ $\quad A u_{\xi\xi} + B u_{\xi\eta} + C u_{\eta\eta} + \dots = 0$ & $\Delta' = (B^2 - 4AC) = \Delta \cdot y^2$. &

$u_{xy} = u_{\xi\xi} \xi_x \xi_y + u_{\xi\eta}(\xi_x \eta_y + \xi_y \eta_x) + u_{\eta\eta} \eta_x \eta_y + \dots$

Unless the coef-s a, b, c = const, the coord. (ξ, η) might be different at every pt in Ω. Also $\Delta > 0$ = hyp, $\Delta = 0$ (par), $\Delta < 0$ Ellip.

Laplace eq-n: $\Delta u = \underline{\nabla}\cdot\underline{\nabla} u = 0 \Rightarrow u_{xx} + u_{yy} + \dots = 0$

Spherical: $u(r,\theta,\varphi)$ $\quad \Delta u = \frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[r^2 \cdot \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \Lambda_3 u = 0$ where oper-r Λ_3 i

$$\Lambda_3 u = \frac{1}{\sin\varphi} \cdot \frac{\partial}{\partial \varphi}\left[\sin\varphi \cdot \frac{\partial u}{\partial \varphi}\right] + \frac{1}{\sin^2(\varphi)} \cdot \frac{\partial^2 u}{\partial \theta^2}$$

Polar: $\quad \Delta u = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \cdot \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \cdot \frac{\partial^2 u}{\partial \theta^2} = 0$

Calc III.

1) Diverg. thm.

$$\underbrace{\int_\Omega \underline{\nabla}\cdot\underline{v}\, d\underline{x}}_{\text{vol. S}} = \underbrace{\int_{\partial\Omega} \underline{v}\cdot\underline{n}\, ds}_{\text{surf. S.}}$$

Ω is regular dom if Ω is open & conn. (dom), bounded, has Lipschitz boundary (finit # of corners)

1) Green's first Id-ty:

$$\int_\Omega u \cdot \underline{\nabla}^2 w\, d\underline{x} = \int_{\partial\Omega} u \cdot \frac{\partial w}{\partial \underline{n}}\, ds - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w\, d\underline{x}$$

(← Integration by parts)

2) Green's 2nd id-y:

$$\int_\Omega (u\, \underline{\nabla}^2 w - w \cdot \underline{\nabla}^2 u)\, d\underline{x} = \int_{\partial\Omega} \left(u \frac{\partial w}{\partial n} - w \frac{\partial u}{\partial \underline{n}}\right) ds.$$

Heat eq-n: $\rho c \cdot u_t - \underline{\nabla}(k \cdot \underline{\nabla} u) = 0 \overset{k = \text{const.}}{\Rightarrow} \frac{\rho c}{k} \cdot u_t - \underline{\nabla}\cdot(\underline{\nabla} u) = 0$ & $t' = \frac{k}{\rho c} t$

$u(\underline{x}, t) = ?$ Addit. info needed: must be given.

IC: $u(x,y,z,0) = \varphi(x,y,z)$ $\forall x,y,z \in \Omega$, $\boxed{u \in C^2(\Omega) \cap C^1(\bar{\Omega})}$

BC: What's happenig on $\partial\Omega$

① Dirichlet BC: $u(x,y,z;t) = f(x,y,z;t)$ $\forall (x,y,z) \in \partial\Omega$, $t \ge 0$

② Neuman BC: $\frac{\partial u}{\partial \underline{n}}(x,y,z;t) = g(x,y,z;t)$ $\forall (x,y,z) \in \partial\Omega$, $t \ge 0$.

③ Mixed $\alpha \frac{\partial u}{\partial \underline{n}} + \beta u = h(x,y,z;t)$ (needed if convective BC)

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§6.5. well-Posed: ① sol-n $\exists$ ② Sol-n unique ③ $\forall |f - \bar{f}(x,y)| < \delta \Rightarrow |u - \bar{u}| < \varepsilon(\delta)$

§7.2 Sep-n of var-s; $u(r,\theta) = R(r) \cdot \Theta(\theta)$

§7.3. Preserving Har-ty: ① Trs $x' = x + x_0$, $y' \dots$, ② Rotations, ③ Reflections $x' = x$, $y' = -y$, ④ Stretch $x' = \lambda x$, $y' \dots$

§7.5. Prop-s of Laplace eq-n $\Delta u = 0$

Prop ② of Har. u: Thm MVP $\quad u(\underline{r}_0) = \frac{1}{|S(\underline{r}_0, \delta)|} \int_{S(\underline{r}_0, \delta)} u(x')\, ds$, $\forall r_0 \in \Omega$. (surface)

Prop ③ of harm. u: Thm (Max. Pr): u attains max/min on $\partial\Omega$.

D'Alambert sol-n $\quad u = F(x+ct) + g(x-ct)$ $\quad u(x,t) = \frac{1}{2}[\varphi(x+ct) + \varphi(x-ct)] + \frac{1}{2}\int_{x-ct}^{x+ct} \psi(\tau)\, d\tau$